

ASSIGNMENT No: 9

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TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives: To implement the final keyword with variables, methods, and classes.

Theory :

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Exercise:

1. Create a Bank Account program where account number is final and cannot be changed.

```
J Bank.java > Bank > main(String[])
1  class BankAccount {
2
3      final int accountNumber = 12345;
4      String accountHolder = "Rahul";
5      double balance = 5000.0;
6
7      void display() {
8          System.out.println("Account Number: " + accountNumber);
9          System.out.println("Account Holder: " + accountHolder);
10         System.out.println("Balance: " + balance);
11     }
12 }
13
14 public class Bank {
15     Run | Debug
16     public static void main(String[] args) {
17         BankAccount b = new BankAccount();
18         b.display();
19     }
20 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.j
Account Number: 12345
Account Holder: Rahul
Balance: 5000.0
PS C:\Users\arhaa\OneDrive\Desktop\java>
```

2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
library.java > library
1  class LibraryBook {
2
3      final String ISBN = "978-81-12345-67-8";
4      String title = "Java Programming";
5      String author = "James Gosling";
6      double price = 450.0;
7
8      void display() {
9          System.out.println("ISBN: " + ISBN);
10         System.out.println("Title: " + title);
11         System.out.println("Author: " + author);
12         System.out.println("Price: Rs." + price);
13     }
14 }
15 public class library{
    Run | Debug
16     public static void main(String[] args) {
17         LibraryBook book = new LibraryBook();
18         book.display();
19     }
20 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Microsoft Visual Studio\2019\Community\VC\Tools\MSVC\14.29.30133\bin\Hostx64-x64\cmd.exe' /s /c cd /d %cd% & java -cp . library
ISBN: 978-81-12345-67-8
Title: Java Programming
Author: James Gosling
Price: Rs.450.0
PS C:\Users\arhaa\OneDrive\Desktop\java>
```